Uncertainty Modeling for Domain Generalization in Cross-Scene Hyperspectral Image Classification

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***Abstract*—Domain generalization is a promising approach for cross-scene hyperspectral image classification (HSIC). Existing domain generalization-based methods usually neglect uncertainty in feature statistics, which may make the trained model vulnerable to statistical changes. Uncertainty modeling may solve this problem by explicitly characterizing the distribution of feature statistics. However, introducing it directly faces two obstacles: properly describing uncertainty is challenging, and the convergence of the entire model is not guaranteed. To address the above issues, in this article, we propose an uncertainty modeling method for domain generalization (UMDG) in cross-scene HSIC, which consists of a newly designed framework and a convergence theorem. The framework develops two strategies to model uncertainty in input and label spaces. In the input space, we propose a variational style augmentation approach to model the explicit distribution of feature statistics across domains by sampling from a learned Gaussian distribution. To maintain consistency, in the label space, we introduce a label mixup strategy that leverages the same uncertainty to increase label diversity. Furthermore, to ensure convergence of the entire model, we establish a convergence theorem by demonstrating that the model converges in expectation to a stationary point. The major contributions of UMDG include designing a UMDG in cross-scene HSIC and providing a convergence theorem. Our experiments on three popular datasets have substantiated the significance of our work.**

***Index Terms*—Domain generalization, hyperspectral image classification (HSIC), uncertainty modeling.**

I. Introduction

HYPERSPECTRAL images, with continuous spectral signatures and rich spatial information, provide valuable insights into the spectral properties of various materials present on the Earth's surface [1], [2], [3], [4]. Their ability to perceive subtle spectral differences makes them widely applicable in various fields [5], [6], [7], [8]. As a pivotal task in hyperspectral data processing, hyperspectral image classification (HSIC) enables the identification of land-cover types within regions of interest [9], [10], [11]. Due to the powerful feature learning capabilities of deep learning, it has become a prevalent approach in HSIC. Commonly employed algorithms include: convolutional neural networks (CNNs), autoencoders (AEs), Transformers, recurrent neural networks (RNNs), and graph convolution networks (GCNs) [12], [13], [14]. CNN-based methods utilize their powerful nonlinear representation abilities to improve classification accuracy [15], [16], [17], [18]. AE-based methods learn a valid encoding of input samples to enhance feature extraction performance [19], [20], [21], [22]. Transformer-based approaches develop various architectures to fully extract spectral–spatial features [13], [23], [24], [25]. RNN-based approaches leverage their proficiency in managing temporal dependencies within hyperspectral data to enhance classification performance [26], [27], [28]. GCN-based methods construct long-range spatial relationships between samples [29], [30], [31]. Part of these methods are based on the empirical risk minimization (ERM) criterion, which assumes that train and test data are independent and identically distributed [32].

One of the challenges in HSIC is the variability in spectral reflectance across scenes of the same land-cover class [33]. This variability can compromise the assumption of training and test data being independent and identically distributed, thereby reducing the effectiveness of the ERM criterion. Furthermore, due to the high labor costs associated with collecting labeled data, we usually hope to use source scenes with sufficient labeled samples to classify unlabeled target scenes [25], i.e., cross-scene HSIC. Popular methodologies for cross-scene HSIC include two main categories: domain adaptation (DA) and domain generalization (DG) [34], [35]. These methods differ based on whether target scene data is utilized during the training phase.

Domain adaptation aims to transfer knowledge from a fully annotated source scene to a target scene with unknown labels by minimizing distributional differences between the scenes [36], [37]. Although domain adaptation methods have demonstrated strong potential, their practical effectiveness may be constrained.

Domain generalization is promising for cross-scene HSIC, especially when the target scene data are challenging to observe [38], [39], [40], [41]. The goal of domain generalization is to develop a model trained on single-source or multisource domains that can generalize effectively to an unseen target scene [42], [43], [44]. For instance, Gao et al. [45] achieved alignment across domains by considering the spectra as a conditional distribution. Zhao et al. [46] proposed a domain expansion network by constructing a local joint feature using unbiased randomization, i.e., LLURNet. Zhang et al.

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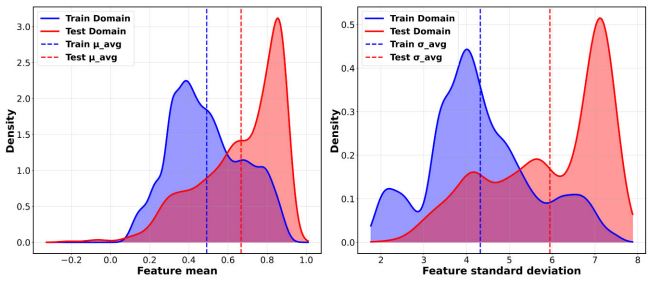


Fig. 1. Quantitative characterization of the uncertain shifts of feature statistics (mean, standard deviation) between the train domain and the unseen test domain via density distributions. Each feature statistic is treated as a probability point with multiple distribution possibilities, rather than a certain value.

[47] developed a novel single-source-domain generalization framework based on the generative adversarial learning, i.e., SDENet. Dong et al. [48] proposed a spectral–spatial enhancement network (S^2ECNet) that enhances generalization ability by learning causal invariances. Qin et al. [49] combined frequency decoupling with data geometry to ensure reasonable expansion distribution, i.e., FDGNet. The existing domain generalization-based methods usually neglect uncertainty in the feature statistics. As shown in Fig. 1, the density distributions of feature statistics (mean and standard deviation) between the training and test domains reveal significant, nonoverlapping discrepancies. This objective fact experimentally verifies the limitation of the ERM criterion, indicating the severe uncertain shifts in feature statistics across scenes. This may make the trained model vulnerable to statistical changes, resulting in limited generalization ability. This observation provides a solid empirical foundation for our proposal to utilize Gaussian distributions to explicitly model this distributional uncertainty. Uncertainty-based domain generalization methods explicitly model the domain shift from the uncertainty perspective, potentially addressing this issue and improving network generalization ability [50], [51], [52], [53]. However, modeling uncertainty for domain generalization in cross-scene HSIC may suffer from two obstacles.

1) Properly describing uncertainty is challenging, as explicit modeling of uncertainty in feature statistics remains unclear.
2) The convergence of the entire model is not guaranteed, which may cause the original classification model to diverge and subsequently affect the optimization.

Existing style-based DG methods rely on heuristic linear interpolation within the source convex hull, struggling with complex out-of-distribution (OOD) shifts. Current uncertainty-aware methods focus only on feature augmentation, risking invalid labels. Our method uses learned Gaussian distributions for OOD feature generation and establishes a consistent uncertainty propagation loop from feature to label space.

In this article, we propose an uncertainty modeling method for domain generalization (UMDG) in cross-scene HSIC, which effectively models uncertainty in input and label spaces with theoretical support. The proposed framework is easily extensible to other HSIC methods. In the input space, to facilitate broader domain transportation, we develop a variational style augmentation approach. This approach randomly samples feature statistics from an estimated Gaussian distribution, replacing the original deterministic values to model uncertainty. In the label space, inspired by the idea of random mixup, we present a label mixup approach that utilizes the uncertainty consistent with feature statistics to further diversify label sources. Furthermore, to guarantee the convergence of the entire model, we establish a convergence theorem by demonstrating that the model converges in expectation to a stationary point. Finally, we integrate the training and deployment processes within a Bayesian meta-learning framework, which is specifically designed for generalizing from a single source. Our proposed method effectively alleviates the performance degradation caused by uncertainty existing in the feature statistics. The key contributions of this work are concisely outlined as follows.

1) We propose a UMDG in cross-scene HSIC, which consists of a newly designed framework and a theoretical support.
2) We propose variational style augmentation and label mixup to model uncertainty in input and label spaces, respectively.
3) We propose a convergence theorem to guarantee the convergence by demonstrating that the model converges in expectation to a stationary point.

II. Methodology

A. Backgrounds

1) Uncertainty Modeling: Uncertainty captures the inherent noise and randomness in data and has received increasing attention in deep representation learning [54]. Some studies proposed leveraging effective representation learning of normality to model uncertainty [54], [55]. This idea can be expressed as

$$p(z_i \mid x_i) = \mathcal{N}\left(z_i; \mu_i; \sigma_i^2 I\right) \tag{1}$$

where z denotes the latent representation, μ is the feature mean, and σ is the standard deviation. Both can be obtained by an encoder. Typically, to make the data sampling operation differentiable, the reparameterization trick is used (see Fig. 2 for an overview pipeline)

$$z_i = \mu_i + \sigma_i \varepsilon \tag{2}$$

here the random noise $\varepsilon \sim \mathcal{N}(0, 1)$.

Unlike traditional Bayesian neural networks (BNNs) focusing on epistemic or aleatoric uncertainty, our work explicitly models distributional uncertainty at the feature level. In cross-scene hyperspectral imaging, physical condition variations cause unpredictable shifts in intermediate feature statistics (mean and standard deviation). Thus, the uncertainty referred to in this article targets these volatile statistical shifts across domains.

2) Bayesian Meta-Learning: Bayesian meta-learning is a probabilistic extension of meta-learning, designed to explicitly model the inherent uncertainty in task distributions and the adaptive process across tasks. The key idea is to introduce a set of task-dependent latent variables ϕ^τ to control the task-specific adaptation process (e.g., the direction and step size of gradient updates), and learn the posterior distribution of these variables through Bayesian inference.

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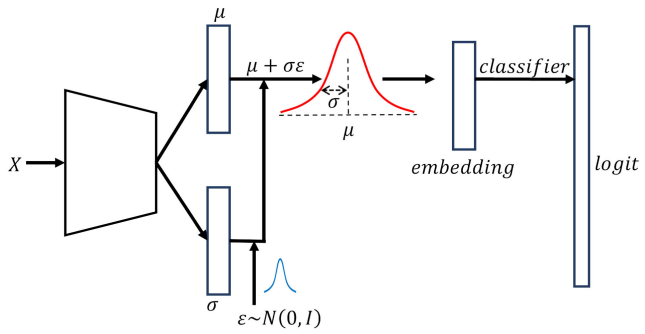


Fig. 2. Overview of the studies that proposed leveraging representation learning of normality to model uncertainty.

In the standard meta-learning setting, the model learns from a task distribution $p(\tau)$, where each task τ includes a training set $D^\tau = (X^\tau, Y^\tau)$ and a test set $\tilde{D}^\tau = (\tilde{X}^\tau, \tilde{Y}^\tau)$. Bayesian meta-learning introduces latent variables ϕ^τ to model the task generation process as

$$p\left(\mathbf{Y}^\tau, \tilde{\mathbf{Y}}^\tau, \phi^\tau \mid \mathbf{X}^\tau, \tilde{\mathbf{X}}^\tau; \theta\right) = p\left(\phi^\tau\right) \prod_{n=1}^{N_\tau} p\left(\mathbf{y}_n^\tau \mid \mathbf{x}_n^\tau, \phi^\tau; \theta\right) \prod_{m=1}^{M_\tau} p\left(\tilde{\mathbf{y}}_m^\tau \mid \tilde{\mathbf{x}}_m^\tau, \phi^\tau; \theta\right) \tag{3}$$

where θ is the shared initialization parameter and $p(\phi^\tau)$ is the prior distribution over the latent variables (typically standard Gaussian). Since the true posterior is intractable, variational inference is employed by introducing an inference network $q(\phi^\tau \mid \mathcal{D}^\tau; \psi)$ to approximate the posterior. The model is trained by optimizing the evidence lower bound (ELBO)

$$\mathcal{L}_{\text{ELBO}}(\boldsymbol{\theta}, \boldsymbol{\psi}) = \mathbb{E}_{q\left(\phi^\tau \mid \mathcal{D}^\tau; \boldsymbol{\psi}\right)}\left[\log p\left(\tilde{\mathbf{Y}}^\tau \mid \tilde{\mathbf{X}}^\tau, \phi^\tau; \boldsymbol{\theta}\right)\right] - \beta \cdot \mathrm{KL}\left[q\left(\phi^\tau \mid \mathcal{D}^\tau; \boldsymbol{\psi}\right) \| \, p\left(\phi^\tau\right)\right]. \tag{4}$$

B. Overall Framework

Our framework, as shown in Fig. 3, integrates an HSIC network with a domain generalization approach. The framework includes a backbone network θ for feature extraction and an auxiliary network $\varphi = \{\varphi_p, \varphi_m\}$. The auxiliary network employs two strategies to address domain shift by modeling uncertainty in input and label spaces. Specifically, φ_p is implemented as two parallel linear layers and φ_m is a single linear layer. φ_p is attached to the intermediate layer of the backbone to project encoded features into statistical spaces (mean and standard deviation), while φ_m acts as a subsequent branch that maps these derived statistics to the corresponding label mixup parameters. This lightweight design ensures efficient uncertainty modeling without introducing excessive computational overhead.

Specifically, the proposed method leverages existing HSI classification methods to extract intermediate feature z. To achieve effective domain generalization, we integrate uncertainty through the auxiliary network $\varphi = \{\varphi_p, \varphi_m\}$, which explicitly models uncertainty in input and label spaces. In the input space, we employ a variational style augmentation strategy to obtain the feature augmentation z^+, which models uncertainty by synthesizing new feature statistics. The uncertainty-driven augmentation strategy can generate OOD samples, thereby enhancing the diversity of the augmented data. Since feature augmentation may introduce uncertainty to the label, in the label space, inspired by random mixup, we further augment the labels with uncertainty encoded in (a, b, ρ). To guarantee the convergence of the entire model, we establish a convergence theorem by demonstrating that the model converges in expectation to a stationary point. To efficiently organize domain augmentation and model training, we jointly optimize the backbone θ and the network φ to maximize the posterior probability of $P(S^+)$ via a Bayesian meta-learning framework.

C. Variational Style Augmentation for Input Space Uncertainty

Considering target scenes with uncertain feature statistics shifts, correctly modeling uncertainty is crucial for addressing domain generalization challenges. The goal of this task is to leverage uncertainty modeling to generate an extended distribution S^+, thereby improving the ability to generalize to unseen scenes. Therefore, the domain shift between S and S^+ should be sufficiently large, ensuring that S^+ can better contain the test set T with uncertainty guarantees. In this article, both the feature statistics μ and σ are obtained in a learnable way, i.e., $(\mu, \sigma) = f_{\varphi_p}(z)$, which are then used to create a broader feature augmentation z^+ by modeling uncertainty in input space.

To ensure the reliability and effectiveness of S^+, we propose learning the encoder φ_p by adversarial domain augmentation as follows:

$$*minimize_{\varphi_p} \mathcal{L}\left(\theta; \mathcal{S}^+\right) - \beta \left\| \mathbf{z} - \mathbf{z}^+ \right\|_2^2 \tag{5}$$

where $\mathcal{L}$ represents the classification loss. $\mathcal{L}$ acts as an implicit semantic constraint, penalizing feature perturbations that corrupt class-discriminative structures. The second term maximizes feature distance for OOD exploration. This adversarial interplay ensures diverse yet semantically reliable S^+. z denotes the output from the fully connected (FC) layer following the convolutional layer. β is a hyperparameter to balance the two terms.

Our style augmentation strategy can effectively relax the constraints and encourage broader domain transportation. Now we can introduce the specific expression of z^+.

The variance of feature statistics carries implicit semantic meaning, and directions with higher variance indicate potentially more meaningful semantic transitions. Consequently, we employ the variance of feature statistics to guide the augmentation process

$$\Sigma_\mu^2(z) = \frac{1}{B} \sum_{b=1}^{B} \left(\mu(z) - \mathbb{E}_b\left[\mu(z)\right]\right)^2 \tag{6}$$

$$\Sigma_\sigma^2(z) = \frac{1}{B} \sum_{b=1}^{B} \left(\sigma(z) - \mathbb{E}_b\left[\sigma(z)\right]\right)^2 \tag{7}$$

where $\Sigma_\mu \in \mathbb{R}^C$ and $\Sigma_\sigma \in \mathbb{R}^C$ denote the standard deviation of the feature statistics μ and σ, respectively.

Unlike nonparameterized mixture methods (e.g., MixStyle) that restrict augmented statistics within the source convex

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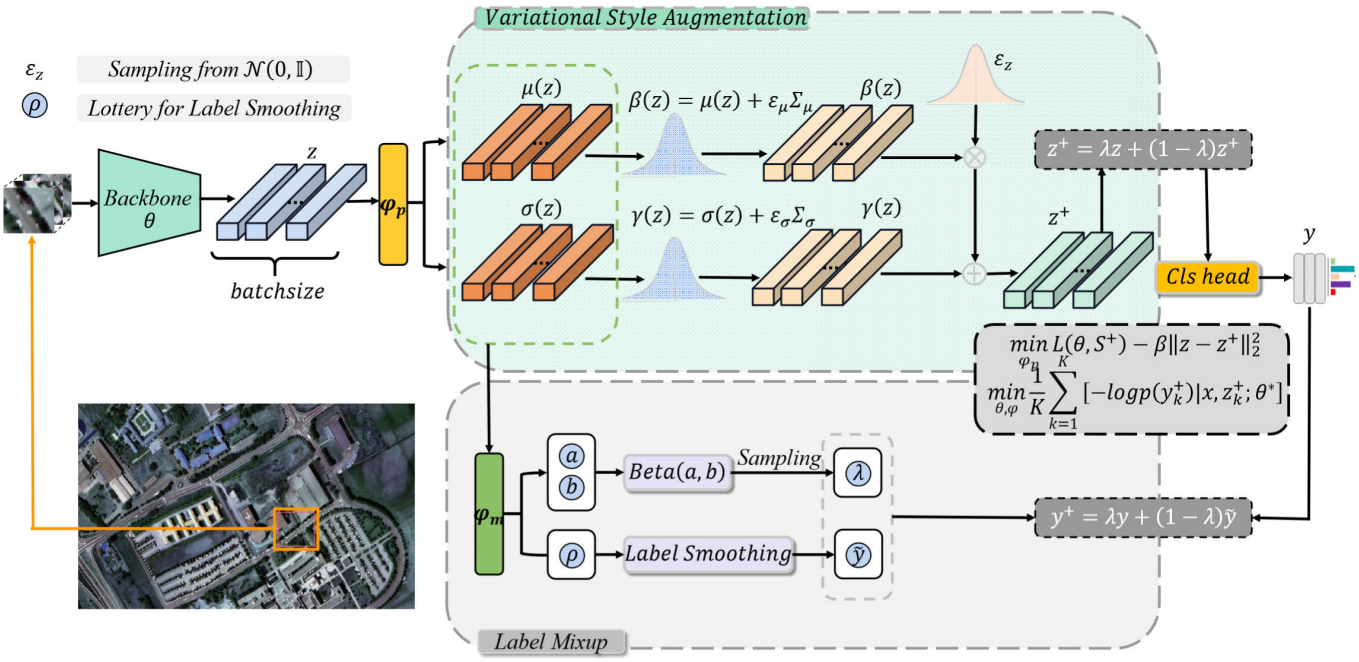


Fig. 3. Flowchart of the proposed UMDG. The overall framework is composed of backbone network for HSIC and two strategies for uncertainty modeling. The first minimization equation is the adversarial domain augmentation objective [see (5)], and the second is the meta-learning objective [see (14)]. All symbols are defined in Section II.

hull via linear interpolation, our variational style augmentation samples from a learned Gaussian distribution. This enables OOD generation and explores diverse direction-intensity combinations. Moreover, the per-channel variance Σ allows adaptive uncertainty modeling across different spectral bands, which is crucial for hyperspectral data.

The standard deviation obtained from the mini-batch reveals the likelihood of potential changes in each channel, providing an appropriate uncertainty estimation for feature statistics. To further incorporate uncertainty into style augmentation, we adopt random sampling to generate new feature statistics. The mean $\beta(z) \sim \mathcal{N}(\mu, \Sigma_\mu^2)$ and standard deviation $\gamma(z) \sim \mathcal{N}(\sigma, \Sigma_\sigma^2)$ are generated as follows:

$$\begin{aligned} \beta(z) &= \mu(z) + \epsilon_\mu \Sigma_\mu(z), \quad \epsilon_\mu \sim \mathcal{N}(\mathbf{0}, \mathbf{1}) \\ \gamma(z) &= \sigma(z) + \epsilon_\sigma \Sigma_\sigma(z), \quad \epsilon_\sigma \sim \mathcal{N}(\mathbf{0}, \mathbf{1}) \end{aligned} \tag{8}$$

where the mean $\mu(z)$ and standard deviation $\Sigma_\mu(z)$ are learned by variational inference $(\mu(z), \Sigma_\mu(z)) = f_{\varphi_p}(z, \theta)$. $z = \theta(S)$ denotes the intermediate feature extracted by the backbone network. By utilizing the Gaussian distribution, we can perform random sampling to produce diverse feature statistics for various direction and intensity combinations. Subsequently, the original feature statistics are replaced with randomly sampled ones. The ultimate expression can be detailed as follows:

$$z^+ = \underbrace{(\sigma(z) + \epsilon_\sigma \Sigma_\sigma(z))}_{\gamma(z)} \epsilon_z + \underbrace{(\mu(z) + \epsilon_\mu \Sigma_\mu(z))}_{\beta(z)} \tag{9}$$

where $\epsilon_z \sim \mathcal{N}(\mathbf{0}, \mathbf{1})$.

D. Label Mixup for Label Space Uncertainty

Since feature augmentation z^+ introduces feature uncertainty, the corresponding label should also reflect this uncertainty. Instead of using the original hard label y, we adopt a learnable label mixup strategy. The parameters (a, b, ρ) are derived from the feature statistic uncertainty φ_m such that the mixing ratio $\lambda \sim \text{Beta}(a, b)$ and the label smoothing factor ρ are adaptive to the uncertainty of the input samples. This ensures consistent uncertainty propagation from the input to the label space.

We introduce φ_m to estimate label uncertainty from the uncertainty in feature statistics μ and σ. The uncertainty is embedded in the parameters (a, b, ρ) for our following label mixup. As shown in Fig. 3, label uncertainty is used to augment the labels, ensuring consistency with the input space. This consistent augmentation further enhances the generalization ability to unseen data. Specifically, φ_m takes the concatenated vector $[\mu, \sigma]$ as input and outputs the parameters (a, b, ρ). The input design is adopted because the mean μ encapsulates the current state of the feature, while the standard deviation σ captures its fluctuation degree. Together, they provide complete guidance for uncertainty injection in the label space.

We begin by introducing random mixup, which is a simple learning principle. The main idea for random mixup is to enhance generalization ability by linearly interpolating between inputs and their corresponding labels. Specifically, two sets of data samples are randomly selected from the training dataset. Linear interpolation is performed on the selected data (x_i, x_j) and their corresponding labels (y_i, y_j) to generate new samples

$$\tilde{x} = \lambda x_i + (1-\lambda) x_j, \quad \tilde{y} = \lambda y_i + (1-\lambda) y_j \tag{10}$$

where λ is a parameter randomly sampled from the Beta distribution.

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Inspired by random mixup, we refine the method to better accommodate the needs of single-source-domain generalization. In particular, we mix the original embedding features z with the enhanced features z^+ and leverage uncertainty in the feature statistics μ and σ to derive the Beta distribution parameters (a, b) for label mixup

$$\mathbf{z}^+ = \lambda z + (1-\lambda) z^+, \quad \mathbf{y}^+ = \lambda \mathbf{y} + (1-\lambda)\tilde{\mathbf{y}} \tag{11}$$

where λ is randomly selected from the Beta distribution and $\tilde{\mathbf{y}}$ is the output after label smoothing.

We use ρ as the label smoothing parameter, where samples are assigned to the true class with probability $\rho \in (0, 1)$ and to any other class with probability $(1-\rho)/(c-1)$. c is the number of classes.

E. Training Strategy Based on Bayesian Meta-Learning

The framework introduces a two-component structure: a backbone network θ and an uncertainty modeling module φ. This leads to a coupled optimization challenge: φ must learn to generate augmentations S^+ that improve the generalization ability of θ, and θ must be robust enough to learn from these augmentations. Simple joint optimization often causes mode collapse: φ may generate noise that destroys θ or trivial perturbations. Bayesian meta-learning decouples the process into inner and outer loops, simulating cross-scene shifts and ensuring challenging yet semantic-preserving augmentations. To address this issue, we employ Bayesian meta-learning, which provides a natural and coherent framework for such bilevel optimization. We maximize the posterior probability of $P(S^+)$ by the following meta-learning procedure, jointly optimizing the backbone θ and the network φ.

The training process consists of an inner loop (meta-train) and an outer loop (meta-test). In the inner loop, θ is updated on the original source data to obtain temporary weights θ^*. In the outer loop, θ^* is evaluated on the augmented data S^+ generated by φ, and the gradients are used to update both θ and φ. This decoupled update ensures stable optimization.

Specifically, we first perform meta-training on the source scene S, followed by meta-testing on the augmented domain S^+ and then meta-update the parameters (θ, φ) using gradients

$$\nabla_{\theta,\varphi} \mathbb{E}_{p(\mathcal{S}^+)} \left[\mathcal{L}\left(\theta^*; \mathcal{S}^+\right) \right], \text{where } \theta^* = \theta - \alpha \nabla_\theta \mathcal{L}\left(\theta; \mathcal{S}\right) \tag{12}$$

where θ^* and α are the meta-trained backbone and the learning rate, respectively.

The goal is to maximize the posterior $p(\mathbf{y}^+ \mid \mathbf{x}, \mathbf{z}^+; \theta^*)$, which is equivalent to maximizing the conditional likelihood $\log p(\mathbf{y}^+ \mid \mathbf{x}, \mathbf{z}^+; \theta^*)$. However, directly solving this requires the true posterior $p(\mathbf{z}^+ \mid \mathbf{x}; \theta^*, \varphi)$, which is often complex to compute. Therefore, we employ an approximate posterior $q(\mathbf{z}^+ \mid \mathbf{x}; \theta^*, \varphi)$ for variational inference, leading to the following approximate lower bound:

$$L_{\theta,\varphi} = \mathbb{E}_{q(\mathbf{z}^+|\mathbf{x};\theta^*,\varphi)} \left[\log \frac{p\left(\mathbf{y}^+ \mid \mathbf{x}, \mathbf{z}^+; \theta^*\right)}{q\left(\mathbf{z}^+ \mid \mathbf{x}; \theta^*, \varphi\right)} \right]. \tag{13}$$

The lower bound $L_{\theta,\psi}$ is maximized by

$$\min_{\theta,\varphi} \frac{1}{K} \sum_{k=1}^{K} \left[-\log p\left(\mathbf{y}_k^+ \mid \mathbf{x}, \mathbf{z}_k^+; \theta^*\right) \right] + \mathrm{KL}\left[q\left(\mathbf{z}^+ \mid \mathbf{x}; \theta^*, \varphi\right) \| p\left(\mathbf{z}^+ \mid \mathbf{x}; \theta^*, \varphi\right) \right] \tag{14}$$

where $\mathrm{z}_k^+ \sim q(\mathrm{z}^+ \mid \mathrm{x}; \theta^*, \varphi)$ and K is the number of Monte Carlo (MC) samples. For the second term, the standard normal distribution $\mathcal{N}(0, \mathbf{I})$ is not suitable for our task, as it may limit the effectiveness of the feature augmentation. Therefore, we use adversarial domain augmentation [see (5)] to train φ_p. The training procedure is summarized in Algorithm 1.

Algorithm 1 Overall Training Procedure

Require: $\mathcal{S}$: Source scene data
Require: K: MC samples
randomly initialized θ, φ
while *not done* **do**
  Meta-train: Compute $\mathcal{L}(\theta; \mathcal{S})$ w.r.t $\mathcal{S}$
  Compute θ^* with gradient descent:
  $\theta^* = \theta - \alpha \nabla_\theta \mathcal{L}(\theta; \mathcal{S})$
  Generate S^+ using variational style augmentation
  Update φ_p using (5)
  for $k = 1, \ldots, K$ **do**
    Generate variational style augmentation $\mathbf{z}_k^+$ using (9)
    Generate $\mathbf{y}_k^+$ using label mixup
    Meta-test: Compute $\mathcal{L}(\theta^*; \mathcal{S}_k^+)$ w.r.t. $\mathcal{S}_k^+$
  end
  Meta-update: Update θ and φ using (14).
end

F. Convergence Theorem Analysis

After integrating an auxiliary network to model uncertainty, the convergence of the entire model is not guaranteed, which may cause the original classification model to diverge and subsequently affect the optimization. Therefore, in this section, we provide a detailed convergence analysis with theoretical support. We first outline the assumptions that underpin the theorem. Next, we present the theorem (Theorem 1), which guarantees the convergence of the optimization.

Assumption 1 (G-Lipschitz Continuity): The loss function $\mathcal{L}$ satisfies G-Lipschitz continuity, i.e., for any $x, y \in \mathcal{X}$, $\|\mathcal{L}(x) - \mathcal{L}(y)\| \leq G\|x - y\|$.

Assumption 2 (L-Smooth): The loss function $\mathcal{L}$ satisfies L-smoothness, i.e., for all $x, y \in \mathcal{X}$, $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$.

Assumption 3 (Boundedness): The loss function $\mathcal{L}$ is upper bounded by M.

Definition 1 ((ϵ, δ)-Stationary Point): For a function $\mathcal{R}(\theta, \varphi)$, a point $(\tilde{\theta}, \tilde{\varphi})$ is called an (ϵ, δ)-stationary point of $\mathcal{R}$ if $\|\nabla_\theta \mathcal{R}(\tilde{\theta}, \tilde{\varphi})\|_2 \leq \epsilon$ and $\mathcal{R}(\tilde{\theta}, \tilde{\varphi}) \geq \max_{\varphi \in \Phi} \mathcal{R}(\tilde{\theta}, \varphi) - \delta$, where $\epsilon, \delta > 0$.

Theorem 1 (The Convergence Theorem): If the loss function in (14) satisfies Assumptions (1)–(3), then after T iterations, the sequence $\{(\theta^t, \varphi^t)\}$ generated by Algorithm (1) converges to a (ϵ, δ) -stationary point. Here, $\epsilon = \mathcal{O}(1/\sqrt{T})$ and $\delta = \mathcal{O}(1/\sqrt{T})$.

Remark: Although these assumptions may not hold globally for deep nonconvex networks, they are locally reasonable under common training techniques.

Proof: Let $\mathcal{R}(\theta, \varphi) = \mathbb{E}_{(x,y)}[\ell(f_\theta(x), y)]$ be our objective function. Consider the following update rule:

$$\theta^{t+1} = \theta^t - \alpha_\theta^t \hat{g}_\theta^t, \quad \varphi^{t+1} = \mathcal{P}_\Phi\left(\varphi^t + \alpha_\varphi^t \hat{g}_\varphi^t\right) \tag{15}$$

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where $\hat{g}_\theta^t$ and $\hat{g}_\varphi^t$ are stochastic gradient estimates and $\mathcal{P}_\Phi$ is the projection operator onto the feasible set Φ. We assume that the stochastic gradients satisfy $\mathbb{E}[\|\hat{g}_\theta^t\|_2^2] \le \hat{G}_\theta^2$ and $\mathbb{E}[\|\hat{g}_\varphi^t\|_2^2] \le \hat{G}_\varphi^2$.

According to Assumption 2 (L-smoothness) and using the properties of the stochastic gradient, the expected descent after one step of θ-update can be bounded as

$$\begin{aligned}&\mathbb{E}\left[\mathcal{R}\left(\theta^{t+1},\varphi^t\right)\right]\\&=\mathcal{R}\left(\theta^t,\varphi^t\right)-\left(\alpha_\theta-\frac{L\alpha_\theta^2}{2}\right)\|\nabla_\theta\mathcal{R}\left(\theta^t,\varphi^t\right)\|^2+\frac{L\alpha_\theta^2\sigma_\theta^2}{2}\end{aligned}\tag{16}$$

where σ_θ^2 bounds the variance of the gradient estimator. Rearranging terms yields

$$\begin{aligned}&\left(\alpha_\theta-\frac{L\alpha_\theta^2}{2}\right)\left\|\nabla_\theta\mathcal{R}\left(\theta^t,\varphi^t\right)\right\|^2\\&\le\mathcal{R}\left(\theta^t,\varphi^t\right)-\mathbb{E}\left[\mathcal{R}\left(\theta^{t+1},\varphi^t\right)\right]+\frac{L\alpha_\theta^2\sigma_\theta^2}{2}.\end{aligned}\tag{17}$$

Summing over $t = 1$ to T and applying the telescoping sum, we obtain

$$\begin{aligned}&\left(\alpha_\theta-\frac{L\alpha_\theta^2}{2}\right)\sum_{t=1}^{T}\mathbb{E}\left[\left\|\nabla_\theta\mathcal{R}\left(\theta^t,\varphi^t\right)\right\|^2\right]\\&\le\sum_{t=1}^{T}\left[\mathcal{R}\left(\theta^t,\varphi^t\right)-\mathbb{E}\left[\mathcal{R}\left(\theta^{t+1},\varphi^t\right)\right]\right]+\frac{TL\alpha_\theta^2\sigma_\theta^2}{2}.\end{aligned}\tag{18}$$

By Assumption 3, let M be the upper bound of the loss, and then,

$$\begin{aligned}&\left(\alpha_\theta-\frac{L\alpha_\theta^2}{2}\right)\sum_{t=1}^{T}\mathbb{E}\left[\left\|\nabla_\theta\mathcal{R}\left(\theta^t,\varphi^t\right)\right\|^2\right]\\&\le\mathcal{R}\left(\theta^1,\varphi^1\right)+M+\frac{TL\alpha_\theta^2\sigma_\theta^2}{2}.\end{aligned}\tag{19}$$

Dividing both sides by $T(\alpha_\theta - \alpha_\theta^2 L/2)$

$$\begin{aligned}&\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\left[\|\nabla_\theta\mathcal{R}\left(\theta^t,\varphi^t\right)\|^2\right]\\&\le\frac{\mathcal{R}\left(\theta^1,\varphi^1\right)+M}{T\left(\alpha_\theta-\frac{L\alpha_\theta^2}{2}\right)}+\frac{L\alpha_\theta^2\sigma_\theta^2}{2\left(\alpha_\theta-\frac{L\alpha_\theta^2}{2}\right)}.\end{aligned}\tag{20}$$

For small α_θ such that $\alpha_\theta - \alpha_\theta^2 L/2 \approx \alpha_\theta$, we have

$$\begin{aligned}&\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\left[\|\nabla_\theta\mathcal{R}\left(\theta^t,\varphi^t\right)\|^2\right]\\&\le\frac{\mathcal{R}\left(\theta^1,\varphi^1\right)+B}{T\alpha_\theta}+\frac{L\alpha_\theta\sigma_\theta^2}{2}.\end{aligned}\tag{21}$$

For any $\varphi \in \Phi$, the progress of the φ-update can be analyzed using the projected gradient descent property. We have

$$\begin{aligned}&\mathbb{E}\left[\mathcal{R}\left(\theta^t,\varphi\right)-\mathcal{R}\left(\theta^t,\varphi^t\right)\right]\\&\le\mathbb{E}\left[\frac{1}{2\alpha_\varphi}\left(\left\|\varphi-\varphi^t\right\|_2^2+\alpha_\varphi^2\hat{G}_\varphi^2-\left\|\varphi-\varphi^{t+1}\right\|_2^2\right)\right].\end{aligned}\tag{22}$$

Summing over $t = 1$ to T and using telescoping sums

$$\begin{aligned}&\sum_{t=1}^{T}\mathbb{E}\left[\mathcal{R}\left(\theta^t,\varphi\right)-\mathcal{R}\left(\theta^t,\varphi^t\right)\right]\\&\le\frac{1}{2\alpha_\varphi}\mathbb{E}\left[\left\|\varphi-\varphi^1\right\|_2^2\right]+\frac{\alpha_\varphi}{2}T\hat{G}_\varphi^2.\end{aligned}\tag{23}$$

We maximize the left-hand side over $\varphi \in \Phi$

$$\begin{aligned}&\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\left[\mathcal{R}\left(\theta^t,\varphi^t\right)\right]\\&\ge\max_{\varphi\in\Phi}\mathbb{E}\left[\mathcal{R}\left(\theta_T,\varphi\right)\right]-\left(\frac{1}{2\alpha_\varphi T}\mathbb{E}\left[\left\|\varphi-\varphi^1\right\|_2^2\right]+\frac{\alpha_\varphi\hat{G}_\varphi^2}{2}\right).\end{aligned}\tag{24}$$

From (21) and (24), we can obtain

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\left[\|\nabla_\theta R\left(\theta^t,\varphi^t\right)\|_2^2\right]\le\mathcal{O}\left(\frac{1}{\sqrt{T}}\right)\tag{25}$$

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\left[R\left(\theta^t,\varphi^t\right)\right]\ge\max_{\varphi\in\Phi}\mathbb{E}\left[R\left(\theta^T,\varphi\right)\right]-\mathcal{O}\left(\frac{1}{\sqrt{T}}\right).\tag{26}$$

Therefore, there exists an iteration $t^* \in \{1, \ldots, T\}$ such that $(\theta^{t^*}, \varphi^{t^*})$ is a (ϵ, δ)-stationary point with $\epsilon = \mathcal{O}(1/\sqrt{T})$ and $\delta = \mathcal{O}(1/\sqrt{T})$, according to Definition 1. This completes the proof. □

III. Experiments

A. Dataset

Pavia Dataset: The Pavia dataset consists of two scenes, Pavia University (PU) and Pavia Center (PC). The dataset was gathered using the reflective optics spectrographic image system (ROSIS), which provides 115 bands with a spectral range of 430–860 nm. The PU dataset contains 103 bands, while the PC dataset contains 102 bands. In this article, 102 spectral bands common to both datasets are used. The PU has 610×340 pixels and the PC has 1096×715 pixels with a spatial resolution of 1.3 m. Seven land-cover classes common to both datasets are included: 1) trees; 2) asphalt; 3) bricks; 4) bitumen; 5) shadow; 6) meadows; and 7) bare soil. The source scene (PU) includes 39 332 samples and the target scene (PC) includes 39 355 samples. Fig. 4 shows their false-color images and ground-truth maps.

Houston Dataset: It includes two distinct scenes: Houston 2013 and Houston 2018. The Houston 2013 dataset, collected by the ITRES CASI-1500 sensor, has a data size of 349×1905 and a spatial resolution of 2.5 m. The dataset comprises 144 spectral bands, with wavelengths ranging from 364 to 1046 nm. The Houston 2018 dataset, with the same wavelength range, has a data size of 2384×601 pixels and a spatial resolution of 0.05 m. It consists of 48 spectral bands. In this article, 48 spectral bands common to both datasets are used, with an overlap region of 209×955 pixels selected. Seven land-cover classes common to both datasets are included: 1) grass healthy; 2) grass stressed; 3) trees; 4) water; 5) residential buildings; 6) nonresidential buildings; and 7) road. The source scene (Houston 2013) includes 2530 samples and the target scene (Houston 2018) includes 53 200 samples. Fig. 5 shows their false-color images and ground-truth maps.

HyRANK Dataset: It consists of two regions: Dioni and Loukia. The Dioni region has a size of 250×1376 pixels, while the Loukia region has a size of 249×945 pixels, both

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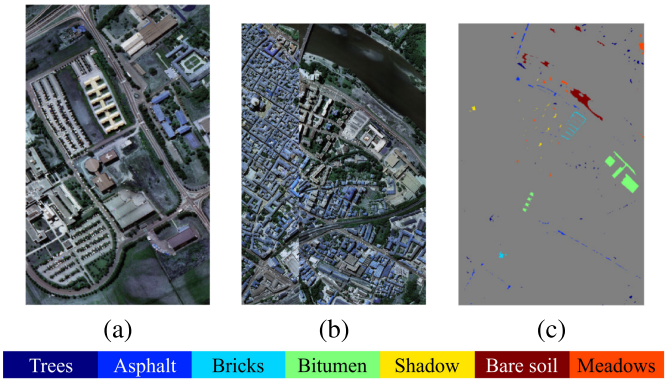


Fig. 4. Pseudo-color image and ground-truth map of Pavia dataset. (a) Pseudo-color image of PU, (b) pseudo-color image of PC, and (c) ground-truth map of PC.

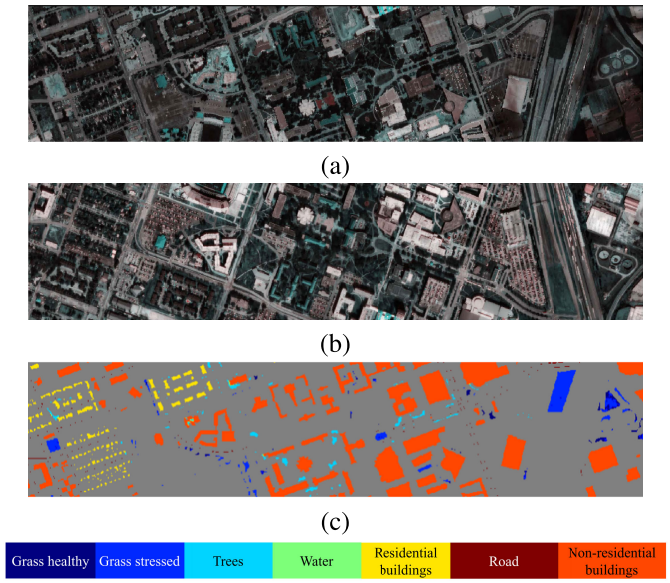


Fig. 5. Pseudo-color image and ground-truth map of Houston dataset. (a) Pseudo-color image of Houston 2013, (b) pseudo-color image of Houston 2018, and (c) ground-truth map of Houston 2018.

with a spatial resolution of 30 m. The data were acquired using the Hyperion sensor, which captures 176 spectral bands and includes 14 consensus land-cover classes. In this article, 176 spectral bands are used. Twelve land-cover classes common to both datasets are included: 1) dense urban fabric; 2) mineral extraction sites; 3) nonirrigated arable land; 4) fruit trees; 5) olive groves; 6) coniferous forest; 7) dense sclerophyllous vegetation; 8) sparse sclerophyllous vegetation; 9) sparsely vegetated areas; 10) rocks and sand; 11) water; and 12) coastal water. The source scene (Dioni) includes 20 024 samples and the target scene (Loukia) includes 10 317 samples. Fig. 6 shows their false-color images and ground-truth maps.

B. Implementation Details

In our study, all experiments are implemented in the PyTorch framework and trained on a single RTX 3090 GPU. The proposed UMDG framework is trained using the Adam optimizer. The batch size is 256, the input patch size is 13×13 pixels, the dimension of the intermediate feature

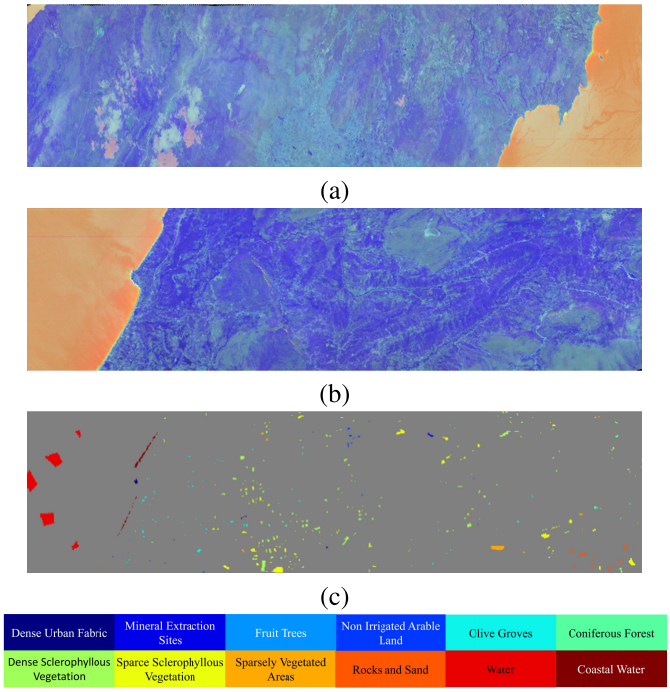


Fig. 6. Pseudo-color image and ground-truth map of HyRANK dataset. (a) Pseudo-color image of Dioni, (b) pseudo-color image of Loukia, and (c) ground-truth map of Loukia.

z is 128, the hyperparameter β in (5) is set to 1, and the number of MC samples K in (14) is 5. For each dataset, 80% of labeled source-domain samples are used for training and the remaining 20% for validation. Each experiment is repeated eight times to ensure reliability. All other backbone networks and comparison methods are configured following their original papers or default parameter settings.

C. Comparison With Plug-and-Play Domain Gneralization Methods

In our experiments, we integrate the proposed method into three advanced hyperspectral classification networks: the spectral classification using a fully convolutional network (SpeFCN) [56], the end-to-end residual spectral–spatial attention network (RSSAN) [57], and the central vector oriented self-similarity network (CVSSN) [17]. To evaluate the performance of the proposed method, we evaluate it in two parts. First, we compare it with the original network, denoted as ERM. Second, we compare it with two plug-and-play domain generalization strategies, Mixstyle [58] and ALOFT [59]. All methods are implemented following the parameter configurations provided in the corresponding studies.

Tables I–III report the results of all methods on the three cross-scene datasets, including class-specific accuracies, overall accuracies (OAs), and Kappa coefficients. As expected, UMDG improves the generalization performance compared with the three backbone networks on the Pavia, Houston, and HyRANK datasets. For SpeFCN, the proposed method outperforms the second-best method by 0.15%, 0.58%, and 0.84% in the OA index for the three datasets. For CVSSN, the performance improvements are 5.45%, 7.30%, and 1.95%. For RSSAN, UMDG achieves the best performance on the HyRANK and Pavia datasets, outperforming the second-best

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TABLE I
CLASSIFICATION ACCURACY OF METHODS ON THE PC DATASET

Class	SpeFcn [56]				RSSAN [57]				CVSSN [17]			
	ERM	MixStyle [58]	ALOFT [59]	UMDG	ERM	MixStyle [58]	ALOFT [59]	UMDG	ERM	MixStyle [58]	ALOFT [59]	UMDG
1	93.18	94.52	91.88	94.60	90.58	97.10	97.00	94.14	96.30	96.68	96.66	95.55
2	62.76	71.89	68.37	74.07	87.69	83.28	73.56	80.98	87.58	86.06	87.44	89.27
3	41.42	23.87	37.73	46.59	38.96	59.52	66.22	31.02	44.99	44.54	44.62	47.41
4	75.15	80.27	90.83	87.13	6.49	27.60	61.23	58.72	35.35	26.76	28.67	63.98
5	82.12	83.27	75.59	81.70	77.09	79.36	80.20	85.47	85.05	85.57	87.81	81.70
6	69.55	71.29	76.70	69.19	74.98	63.82	65.73	70.39	65.28	65.05	69.51	68.58
7	70.26	83.54	77.64	68.74	71.90	70.41	69.29	87.90	59.45	63.46	57.65	57.70
OA(%)	72.67	77.26	77.71	**77.86**	65.48	70.05	74.46	**76.57**	70.04	68.83	68.84	**75.49**
Kappa	67.13	72.45	73.13	**73.35**	58.00	63.96	69.25	**71.68**	64.01	62.62	62.59	**70.51**

TABLE II
CLASSIFICATION ACCURACY OF METHODS ON THE HOUSTON 2018 DATASET

Class	SpeFcn				RSSAN				CVSSN			
	ERM	MixStyle	ALOFT	UMDG	ERM	MixStyle	ALOFT	UMDG	ERM	MixStyle	ALOFT	UMDG
1	28.75	61.27	45.31	29.42	9.02	33.48	38.65	16.85	20.40	18.48	18.26	51.96
2	64.75	27.25	17.57	70.46	65.96	70.62	71.89	73.79	74.75	78.13	73.94	79.50
3	65.04	28.85	15.58	67.03	53.94	41.47	47.76	55.21	34.45	33.19	30.22	52.31
4	81.82	77.27	77.27	86.36	100.00	90.91	81.82	95.45	72.73	72.73	72.73	45.45
5	77.71	51.90	47.50	72.53	50.61	60.05	50.70	46.47	34.58	36.60	32.07	69.63
6	87.87	95.77	95.04	88.27	88.45	94.48	93.95	92.34	98.02	97.80	98.40	94.74
7	12.11	11.23	1.04	13.92	17.39	3.74	9.05	13.48	11.34	13.53	11.55	45.95
OA(%)	72.97	70.59	68.89	**73.55**	70.27	**73.66**	73.60	72.74	73.84	74.37	73.50	**81.67**
Kappa	52.18	39.58	34.98	**53.20**	45.94	49.13	**49.55**	48.44	46.11	47.74	44.70	**67.38**

TABLE III
CLASSIFICATION ACCURACY OF METHODS ON THE LOUKIA DATASET

Class	SpeFcn				RSSAN				CVSSN			
	ERM	MixStyle	ALOFT	UMDG	ERM	MixStyle	ALOFT	UMDG	ERM	MixStyle	ALOFT	UMDG
1	38.35	35.92	3.88	19.90	0.49	29.61	28.16	26.70	38.83	29.13	40.78	34.95
2	0.00	0.00	0.00	0.00	24.07	25.93	64.81	22.22	40.74	59.26	40.74	14.81
3	55.63	44.37	42.72	67.84	23.24	52.82	60.33	52.58	81.22	85.45	78.17	88.03
4	43.04	27.85	5.06	40.51	0.00	0.00	0.00	6.33	79.75	77.22	81.01	48.10
5	3.16	3.61	5.06	14.45	0.09	0.00	0.00	0.27	0.00	0.00	0.00	1.81
6	38.39	39.10	44.31	19.67	42.18	36.49	44.08	15.64	36.97	41.23	32.46	15.64
7	66.12	61.11	61.85	64.49	65.12	69.19	66.62	72.30	70.56	67.82	73.26	82.81
8	51.16	61.37	62.18	58.11	55.65	40.45	55.32	63.66	43.71	43.24	40.87	39.35
9	74.69	78.95	81.45	80.45	36.09	27.57	24.81	19.05	27.07	29.82	30.83	30.83
10	2.65	13.47	0.66	0.00	0.22	19.87	0.22	1.99	0.00	0.00	0.88	2.21
11	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
12	98.57	98.81	98.34	99.52	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
OA(%)	56.74	57.72	57.12	**58.56**	53.47	53.27	55.75	**57.51**	55.59	55.03	55.65	**57.54**
Kappa	48.21	48.75	47.61	**49.81**	42.81	43.08	45.68	**46.91**	47.12	46.82	47.16	**48.11**

method by 1.76% and 2.11%, respectively. On the Houston dataset, it obtains a suboptimal ranking, with its performance being only minimally lower than the top-performing method.

However, an intuitive observation is that Mixstyle and ALOFT perform worse than ERM in some cases. For example, for the backbone CVSSN, Mixstyle and ALOFT perform 1.21% and 1.20% worse than ERM, respectively, on the Pavia dataset. In comparison, UMDG performs relatively satisfactorily on each backbone network. This is because in cross-scene hyperspectral classification tasks, excessive emphasis on randomized sample styles or only performing sample augmentation by interpolation in the feature space may introduce undesired extended distributions, resulting in poor model generalization performance.

Beyond the quantitative evaluation, Figs. 7–9 present the classification maps of different methods. Taking the Pavia dataset as an example, in challenging boundary transition areas (such as between "Bare Soil" and "Meadows"), baseline methods like MixStyle and ALOFT exhibit noticeable salt-and-pepper noise and misclassified patches. In contrast, UMDG produces smoother and more coherent region boundaries with significantly fewer misclassifications. This visual improvement demonstrates that our uncertainty-guided augmentation effectively prevents the model from overfitting to source-domain spectral artifacts. By explicitly modeling and exploring stylistic shifts, the network is forced to rely on essential semantic features rather than domain-specific noise, thereby maintaining high classification consistency even under complex target scene variations.

Some class-specific accuracy fluctuations can be observed in Tables I–III (e.g., Class 3 in the Pavia dataset). This is mainly due to the inherent spectral similarity between certain land-cover classes. When the variational style augmentation generates highly diverse feature statistics to explore broader

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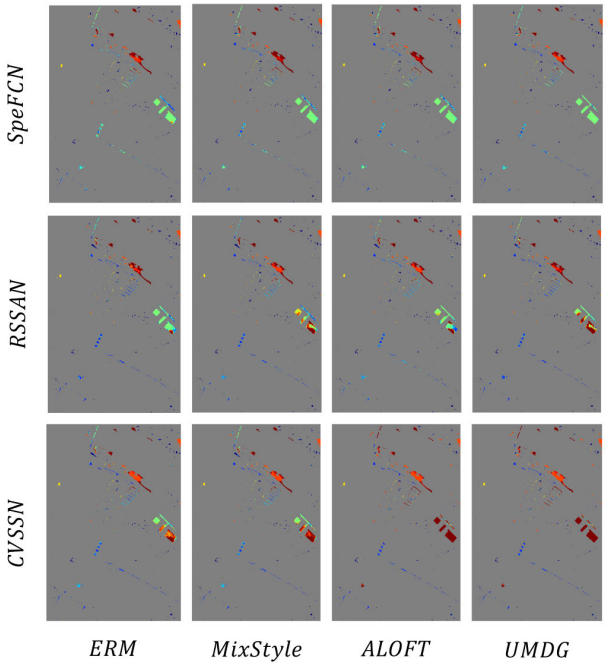


Fig. 7. Visualization and classification maps of different models on the PC dataset.

TABLE IV
COMPARISON WITH DIFFERENT METHODS ON PAVIA DATASET

Class	SDENet	LLURNet	S2ECNet	FDGNet	UMDG
1	83.14	87.22	94.88	79.43	80.89
2	87.07	82.01	80.82	82.66	86.07
3	48.45	73.22	31.69	67.82	78.58
4	76.95	78.03	86.48	82.39	85.52
5	95.67	85.04	93.43	85.33	78.94
6	82.20	74.49	74.17	68.58	77.12
7	70.37	75.18	87.23	68.86	86.06
OA(%)	79.25	80.07	82.70	80.40	**83.24**
Kappa	75.08	76.21	79.06	73.38	**79.90**

domain shifts, the augmented features of a specific class may occasionally cross the decision boundaries of highly similar neighboring classes, causing spectral confusion. While this expansive exploration enhances overall generalization across scenes, it may introduce localized fluctuations for these spectrally adjacent categories.

D. Comparison With Cross-Scene HSIC Methods

In this section, we compare the proposed methods with some state-of-the-art cross-scene HSIC methods, including single-source-domain expansion network (SDENet) [47], locally linear unbiased randomization network (LLURNet) [46], spectral–spatial enhancement and causal constraint network (S^2ECNet) [48], and frequency disentanglement and data geometry network (FDGNet) [49]. Specifically, we incorporate our novel approach into a simple network, which consists of two sets of sequential Conv2d-ReLU-MaxPool2d blocks and three FC layers.

Tables V and VI report the results of all methods on the three cross-scene datasets, including class-specific accuracies, OAs, and Kappa coefficients. Results show that UMDG achieves the best performance on the HyRANK and Pavia datasets and obtains a suboptimal ranking on the Houston dataset. For the Pavia dataset, UMDG improves performance

TABLE V
COMPARISON WITH DIFFERENT METHODS ON THE HYRANK DATASET

Class	SDENet [47]	LLURNet [46]	S2ECNet [48]	FDGNet [49]	UMDG
1	34.47	47.09	52.43	55.83	15.53
2	1.85	0.00	81.48	3.70	87.04
3	41.78	57.04	64.08	48.36	55.40
4	6.33	3.80	8.86	17.72	34.18
5	0.00	0.18	0.45	0.00	3.61
6	40.28	24.17	39.57	30.57	44.08
7	74.77	73.56	74.80	75.70	70.23
8	63.45	72.55	55.44	58.20	70.31
9	24.81	22.31	83.46	30.83	66.67
10	57.84	0.00	0.00	2.87	2.43
11	99.93	100.00	100.00	100.00	100.00
12	100.00	100.00	100.00	100.00	100.00
OA(%)	61.42	60.74	61.07	61.49	**62.26**
Kappa	52.26	51.22	52.63	49.35	**53.81**

TABLE VI
COMPARISON WITH DIFFERENT METHODS ON THE HOUSTON DATASET

Class	SDENet	LLURNet	S2ECNet	FDGNet	UMDG
1	46.05	21.80	92.31	25.72	23.28
2	61.89	78.05	60.66	76.74	46.48
3	60.85	56.76	37.53	59.26	61.28
4	100.00	100.00	100.00	100.00	100.00
5	80.61	79.82	74.19	75.97	68.66
6	86.42	85.05	73.87	90.52	90.69
7	42.48	39.54	80.20	40.44	47.29
OA(%)	75.98	75.36	72.04	**78.53**	75.98
Kappa	58.83	57.94	57.37	**61.24**	57.31

TABLE VII
ABLATION STUDY OF VARIATIONAL STYLE AUGMENTATION

	paviaC_Spe	Houston18_CVSSN
Full Model	77.86	81.67
Random	73.57	79.91
Uniform	72.21	76.36

by 0.54%, while the second-best model S^2ECNet requires significantly longer training time due to the causal inference mechanism. For the Houston dataset, the OA results of UMDG are lower than those of FDGNet, but UMDG demonstrates superior class-specific performance for a large proportion. The performance of UMDG varies across datasets. In Houston, the overall accuracy of UMDG is slightly below that of FDGNet. Unlike Pavia and HyRANK with macro-level covers, Houston contains fine-grained urban classes with narrow boundaries. The aggressive OOD exploration of UMDG, while effective for large domain shifts, may occasionally cross these narrow boundaries, causing localized confusion. This observation highlights a tradeoff between broad distribution exploration and fine-grained structural preservation.

E. Ablation Experiment

In this section, we perform an ablation study of the proposed method on the Pavia and Houston datasets. Specifically, the model is trained using SpeFCN on the Pavia dataset and CVSSN on the Houston dataset. We report the OAs in Tables VII–IX.

Variational Style Augmentation: We investigate different designs of style augmentation: *1) random Gaussian:* feature statistics $\gamma(x), \beta(x)$ are sampled from $\mathcal{N}(0, I)$ with no learnable parameters and *2) uniform distribution:* the shift is extracted from $U(-\Sigma, \Sigma)$, where Σ is the range obtained by modeling

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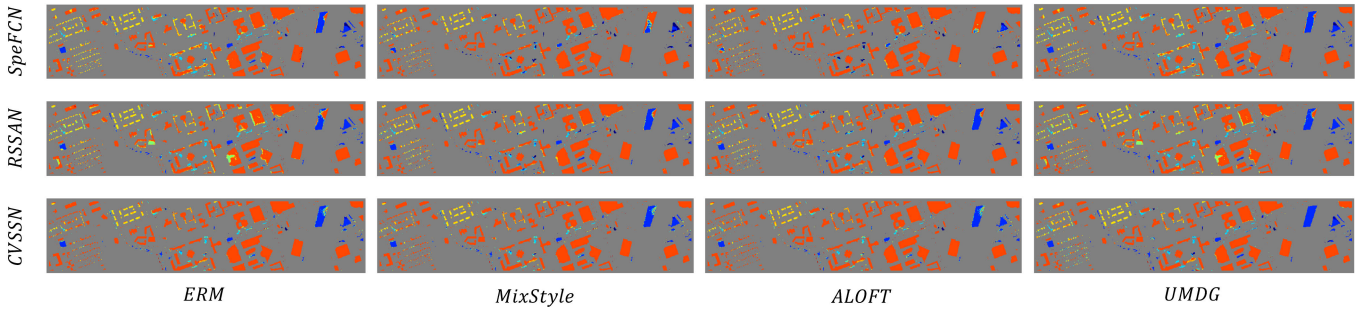


Fig. 8. Visualization and classification maps of different models on the Houston 2018 dataset.

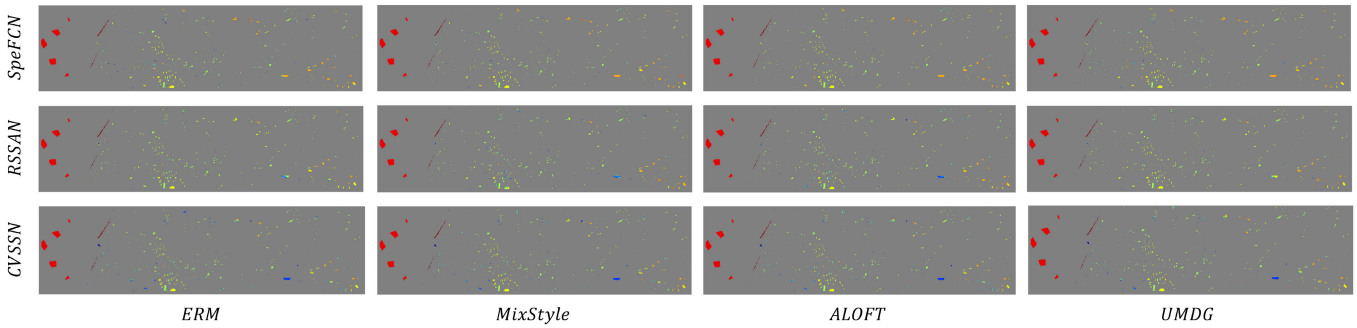


Fig. 9. Visualization and classification maps of different models on the Loukia dataset.

TABLE VIII
ABLATION STUDY OF LABEL MIXUP

	paviaC_Spe	Houston18_CVSSN
Full Model	77.86	81.67
w/o mixup	71.57	79.88
Random mixup	72.72	79.88

TABLE IX
ABLATION STUDY OF TRAINING STRATEGY

	paviaC_Spe	Houston18_CVSSN
Full Model	77.86	81.67
w/o adversarial training	75.11	78.26
w/o meta-learning	74.70	78.11

uncertainty. The results for these different choices are shown in Table VII. From the results, it can be seen that the uniform distribution has the lowest accuracy on both datasets, indicating that the Gaussian distribution is more suitable for modeling the uncertainty of the feature statistics. Additionally, employing a Gaussian distribution with an inappropriate variance range may adversely affect model performance. This implies that the variance range of feature statistics is instructive.

Label Mixup: We conduct experiments with two label mixup variants: *1) without mixup*—the model is only trained with feature augmentation and *2) random mixup*—the mixup parameter λ is randomly selected from a Beta(1,1) distribution. The results for these approaches are shown in Table VIII. We observe that random mixup outperforms the approach without mixup. The data from the experiments demonstrate that incorporating label augmentation techniques can significantly enhance the model's generalization ability. The learnable mixture model (full model) has the best performance, indicating that label augmentation can increase the diversity of sources, further improving the generalization ability of the model.

Training Strategy: Finally, we present a comparison of different training strategies: *1) without adversarial training*—the model is trained without adversarial training as described in (5) and *2) without meta-learning*—the source S and augmentations S^+ are trained together without the meta-learning scheme. The results are reported in Table IX. Meta-learning contributes the most, with a 3.16% improvement on the Pavia dataset and a 3.56% improvement on the Houston dataset. Adversarial training consistently improves accuracy on both datasets. The performance drop without meta-learning stems mainly from gradient conflicts and poor OOD sample absorption. Joint training yields conflicting gradient directions from clean and augmented data, causing suboptimal minima. Meta-learning aligns gradients via its inner and outer loops, enabling smooth extraction of generalizable features from extreme OOD samples without forgetting source knowledge.

IV. CONCLUSION

In this article, we propose a UMDG in cross-scene HSIC, which consists of a newly designed framework and a convergence theorem. The proposed method improves the generalization ability of the network by explicitly modeling domain shift with uncertainty. To ensure the convergence of the entire model, we establish a convergence theorem by demonstrating that the model converges in expectation to a stationary point. The major contributions of UMDG include designing a new framework for domain generalization in cross-scene HSIC with two strategies and providing a convergence theorem. Experiments on the Pavia dataset, Houston dataset, and HyRANK dataset validate the effectiveness of UMDG.

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